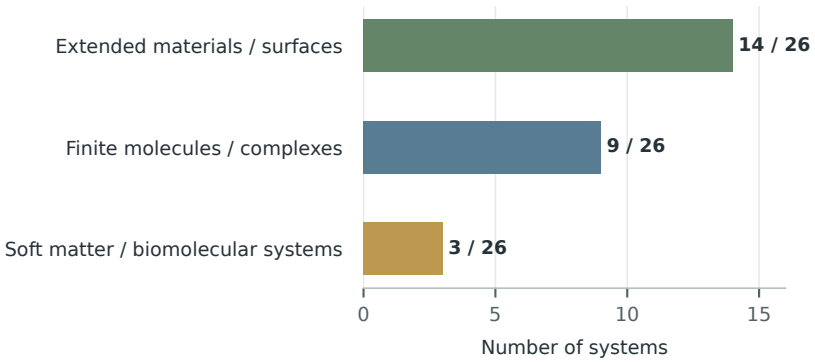
Computational chemistry and atomistic simulation agents

Structured literature sample · 2023–26 October 2026 · 26 reported systems

a Demonstrated physical targets



Physical target ≠ boundary condition

A protein or polymer in a periodic cell remains a biomolecular or polymer target. Software support is not a demonstrated use case.

b Tasks demonstrated in the inspected primary reports

Filled squares: positive evidence. Blank cells: unencoded, not evidence of missing capability. Coverage is not task difficulty, end-to-end completion or scientific acceptance.

	Build / edit structures n = 18	Energy / properties n = 17	Geometry relaxation n = 18	Freq. / thermochem. n = 9	Transition paths / TS n = 6	MD / quantum dynamics n = 8
LLaMP	■					■
ChemReasoner	■	■	■			
AtomAgents	■	■	■		■	
Adsorb-Agent	■	■	■			
MDAgent						■
DREAMS	■	■	■			
VASPilot		■	■			
MASTER	■	■	■			
Masgent	■	■	■			
MatSciAgent	■					■
TSAgent		■	■	■	■	
CatDT	■	■	■	■	■	
AdsMind	■	■	■			
Agent-MD						■
El Agente Q	■	■	■	■		
Aitomia	■	■	■	■		
ChemGraph	■	■	■	■		
El Agente Estructural	■		■			
El Agente Quntur	■	■	■	■	■	
El Agente Forjador	■	■	■	■		■
Evolving TS workflows			■	■	■	
ARCHE	■	■	■	■	■	
Multireference QC agent		■				
MDCrow						■
PolyJarvis	■					■
MDForge						■

Cloudra · this work

Organic small-molecule mechanism exploration

Chemistry-aware perception + geometry operators support geometric reasoning; an evidence DAG carries scientific validation. Restraint and guided diagnosis connect these designs. Cloudra is shown as a design focus and is excluded from literature counts.

Scope: task execution, planning, code modification or feedback decisions involving actual atomistic calculations.
Counts describe this screened sample, not field prevalence; tasks overlap. Preprint and journal versions are deduplicated.
Source-coded records, exclusions and sensitivity analyses accompany the figure.